

✓ Fibonacci Number

Question

The Fibonacci numbers, commonly denoted $F(n)$ form a sequence, called the Fibonacci sequence, such that each number is the sum of the two preceding ones, starting from 0 and 1.

That is,

$$F(0) = 0, F(1) = 1$$

$$F(n) = F(n - 1) + F(n - 2), \text{ for } n > 1.$$

Given n , calculate $F(n)$.

Constraints

$$0 \leq n \leq 30$$

✓ Solution

for doing this question you have to know about how recursion works properly... here i do it by recursion method.

Now for recursion we need a base condition and here base condition is given as --

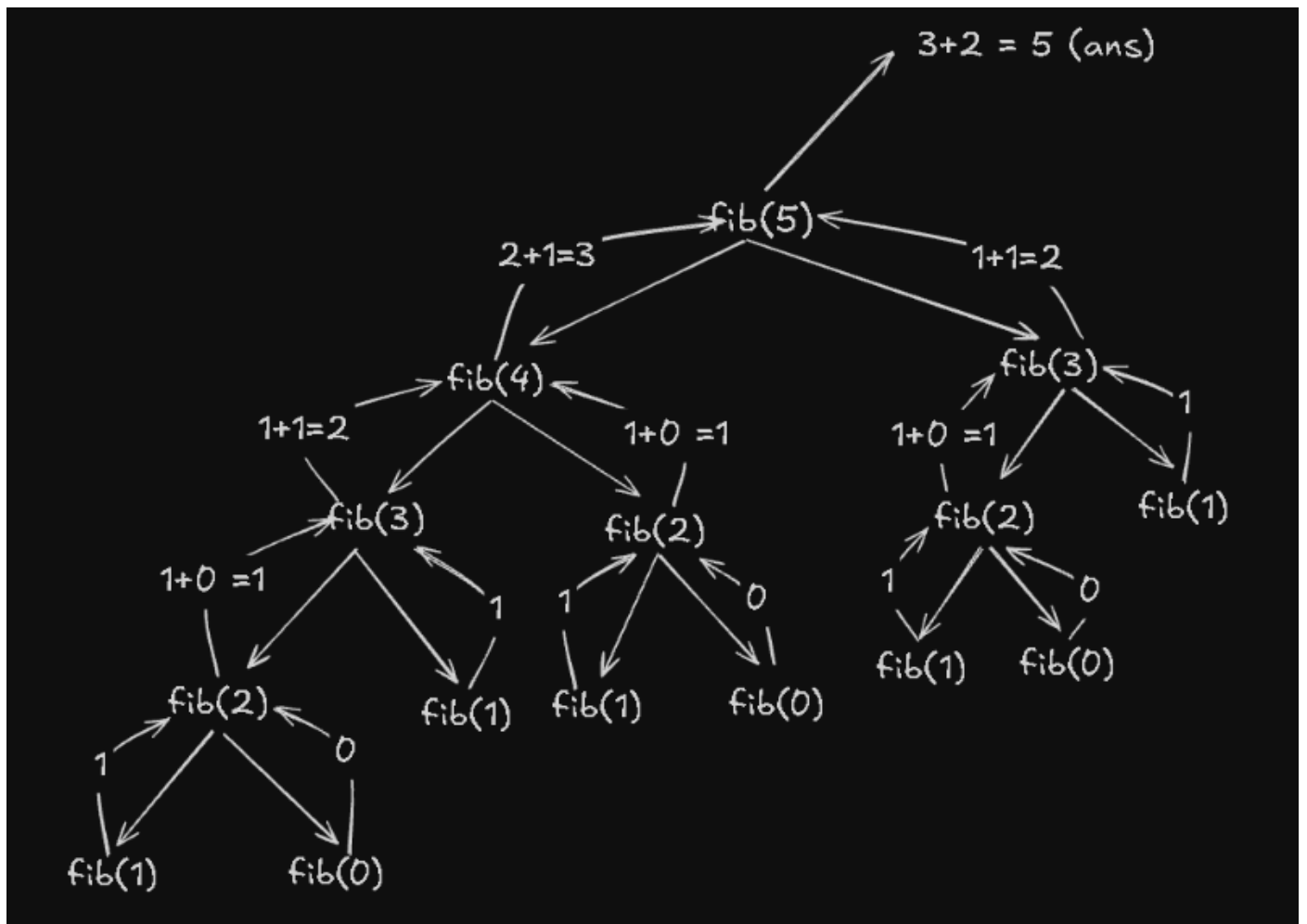
- $F(0) = 0$
 - $F(1) = 1$
-

Pseudo code

```
fib(n){
  if(n==0 or n==1){
    return n
  }
  else{
    return fib(n-1)+fib(n-2)
  }
}
```



Recursion Tree (n = 5)



Code

```
class Solution {
    public int fib(int n) {
        if(n==0 || n==1){    // java or operator --> ||
            return n;
        }else{
            return fib(n-1)+fib(n-2);
        }
    }
}
```

time complexity = $O(2^n)$

space complexity = $O(N)$

